King Abdulaziz University

College of Engineering

Electrical and Computer Engineering

EE 424 (Antenna)

Assignment L15

(Printed Dipole Antenna)

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Reequipments: Design and simulate a printed antenna at 1.2GHz, using Rogers RT/duroid 5880 as a substrate with 2.2 dielectric constant and Hight with 0.5mm.

First, we start with calculating dimensions of printed dipole using formulas we studied in L15:

Lamda= 250mm

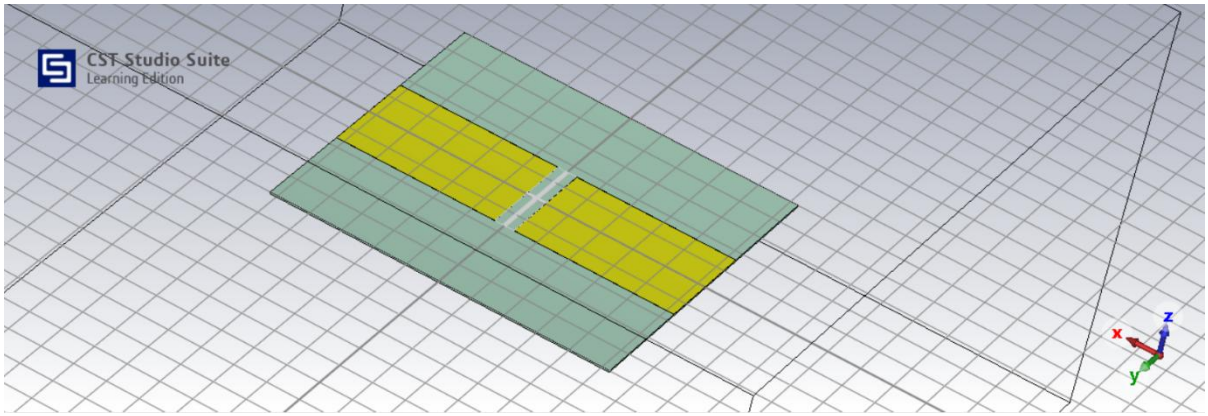
Effective wavelength= 197.211mm

Width = 32.868mm

Length = 98.605mm

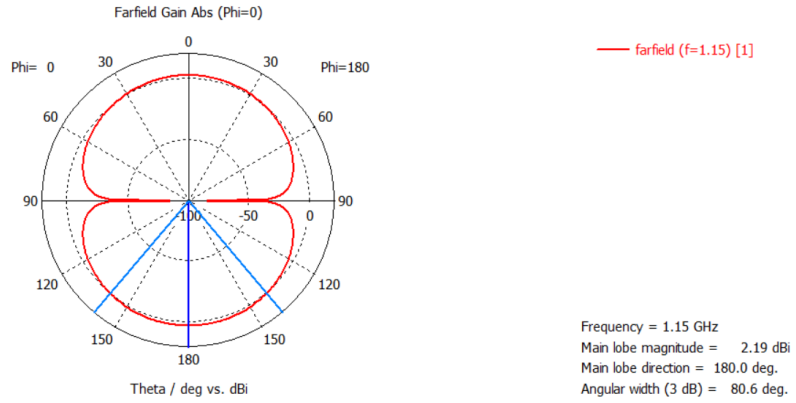
Effective dielectric constant = 1.607

All the calculations performed by hand on last page**

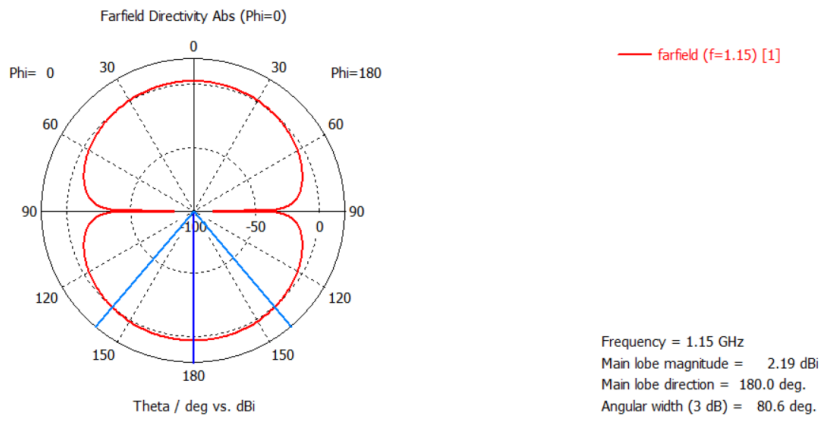


Printed dipole

We can see the gain and directivity of 2.19 dBi with high beamwidth 80.6 deg. Gain and directivity is almost same as the normal dipole was 2.09 dBi.

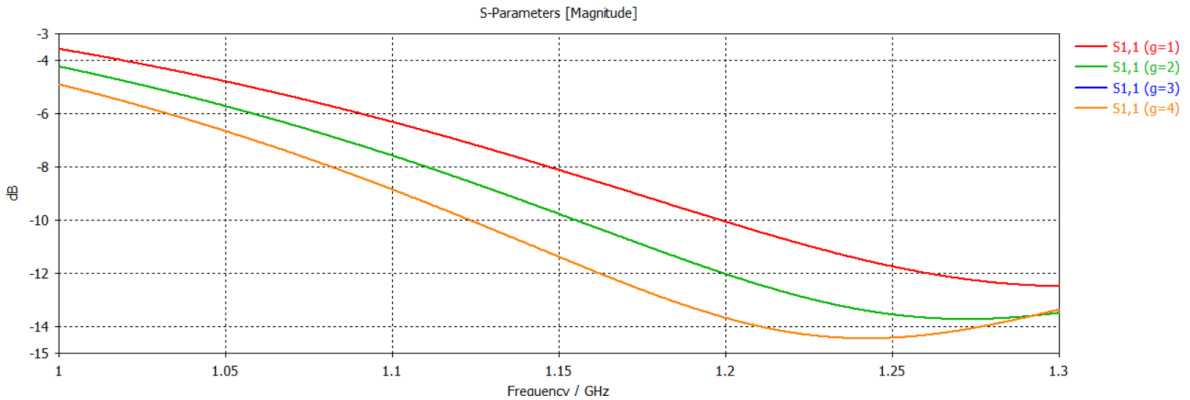


Gain



Directivity

Doing more study on printed dipole antenna I figure out that gab between poles can affect S11 the higher the gab the better S11 get. But there's a limit and it was 3mm it will not get better than that.



S11

References:

1- Lecture 15

2-Calculations done by hand:

L 15

$\varepsilon_r = 2.2$ $\quad h = 0.5\,mm$ $\quad f = 1.2\,GHz$ $\quad c = 3\times10^8$

$\lambda = \frac{c}{f} = \frac{3\times10^8}{1.2\times10^9} = 250\,mm$ $\qquad \lambda_e = \frac{\lambda_0}{\sqrt{\varepsilon_e}}$

$W = \frac{L}{3} = \frac{\lambda_e}{6} = \frac{\lambda_0}{6\sqrt{\varepsilon_e}}$

$L = \frac{\lambda_e}{2}$

$\varepsilon_e = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left[\frac{1}{\sqrt{1+12\left(\frac{h}{2}\cdot 6\sqrt{\varepsilon_e}\right)}}\right]$

$\varepsilon_e = 1.607$

$W = \frac{250\,mm}{6\sqrt{1.607}} = 32.868\,mm \;,\; 25.928\,mm$

$L = W\times3 = 98.605\,mm \;,\; 77.78\,mm$

$\lambda_e = \frac{c}{f\sqrt{\varepsilon_e}} = 197.211\,mm$

$Gap = 1\,mm$